

MAT892 Learning Goals

1. Introduction

- Know the axioms of discrete probability and the formal objects involved.
- Understand the possible worlds interpretation of discrete probability.
- Be able to explain how probability can be used to express beliefs.
- Know the conditioning formula and understand how it relates to updating beliefs.
- Understand the idea behind Bayesian inference (generative model + conditioning).
- Be familiar with the bolded terms in Section 1.1.5.
- Be able to perform Bayesian inference using forward simulation and rejection sampling.

2. Discrete Graphical Models

- Know the definition of a discrete (probability) kernel.
- Be able to interpret kernels.
- Know the formulas for sequential, parallel and copy-composition of discrete kernels.
- Be able to draw and interpret the elements of string diagrams (boxes, wires, bundling, crossing, copy, delete).
- Understand the meaning of boxes with no inputs or outputs.
- Know how to lift functions to deterministic kernels.
- Be able to parse a given string diagram in at least one valid way.
- Understand how the operations on kernels can be represented using tensors (2.2.3).
- Be able to perform forward simulation from a string diagram of probability kernels.
- Be able to estimate expectations and probabilities using the Law of Large Numbers (2.3.3).

3. Conditioning

- Understand the graphical procedure for conditioning.
- Be able to use the graphical test for conditional independence.
- Understand how kernels can be factored using weights (3.3.1) and how this gives rise to importance sampling.
- Know how to sample from diagrams of probability kernels and weights (3.3.3) and understand how the technique applies to Bayesian inference.

4. Hybrid Models

- Know the definition of measurable spaces and measures.
- Know the general definition of kernels (4.1.4).
- Know the definition of density (4.1.5).
- Understand how the definition given for discrete kernels relates to densities.
- Know the formulas for sequential, parallel and copy-composition of kernels with densities.
- Understand how the graphical conditioning procedure works on kernels with densities.
- Understand the need for parametric kernels and how to construct them from base families and deterministic transforms.
- Be familiar with the parametric forms of linear, logistic, and Poisson regression. Understand how they arise as a composition of base families and deterministic transforms.

5. Bayesian Data Analysis

- Internalize the main stages of the Bayesian workflow and know why they are important.
- Be able to read the synthetic notation for model specification ($x \sim \text{Normal}(\dots)$) and know how it translates to NumPyro.
- Understand plate notation.
- Know the basic NumPyro modelling constructs (`.sample` and `.deterministic`) and understand what they do.
- Know what to consider when choosing a prior.
- Know what NumPyro construct to use for prior predictive checks.
- Understand the options you can set in NumPyro's MCMC (`kernel`, `num_warmup`, `num_samples`, `num_chains`) and what inference returns.
- Be able to interpret the ArviZ summary output (`mean`, `sd`, `ess`, `r_hat`).
- Understand how to interpret trace plots and know about the existence of rank plots.
- Be able to interpret pair plots of the posterior (5.7.1).
- Understand how to compute a derived quantity from posterior samples.
- Understand the purpose of posterior predictive checks and how to generate posterior samples in NumPyro.
- Understand how to use summary statistics for posterior predictive checks.
- Know what sensitivity analysis is, how you would do it naively, and why it is important.
- Know how to compare models using the expected log predictive density.
- Understand the difference between out-of-sample prediction, cross-validation, and in-sample prediction.

Not examinable:

- The workings of power-scaling sensitivity analysis.
- The PSIS method for model comparison.

6. Mechanics of MCMC

- Understand the concepts of Markov chain, transition kernel and stationary distribution.
- Understand the difference between spatial and time averages.
- Understand the statement of the ergodic theorem.
- Be able to name two obstructions to the ergodic theorem.

- Be able to intuitively explain Harris recurrence and aperiodicity (exact definitions are not required, only what they convey).
- Understand that Harris recurrence guarantees the ergodic theorem and that added aperiodicity guarantees convergence to the stationary distribution from any initial distribution.
- Know the detailed balance condition and understand intuitively why it implies stationarity.
- Know how to construct a Metropolis–Hastings kernel from a given proposal (including the acceptance formula).
- Know the MH algorithm.
- Understand how MH applies to the Bayesian inference problem.
- Know sufficient conditions for the MH kernel to be aperiodic and Harris recurrent.
- Understand the role played by the proposal in the performance of MCMC.
- Know three types of proposal mechanisms (independent, random-walk, gradient-based) and understand their pros and cons.

Not examinable:

- Section 6.4.8 onwards (apart from the MH proposals).

7. Inference in Practice

- Understand how to marginalize discrete latents.
- Understand MAP and Laplace approximation, including when they are appropriate and their limitations.
- Understand how to exploit conditioning to split models into independent parts, including empirical Bayes.
- Know how to deal with missing data (both when missingness is informative and uninformative) and how this relates to imputation.

Not examinable:

- The details of conjugate distributions.
- How to code models with missing data.

8. Sequential Monte Carlo

- Know the definition of a state-space model.
- Understand the filtering problem.
- Understand how to derive a weighted sampling algorithm (8.3.1) from the SSM model.
- Know the definition of ESS and why it is useful.
- Understand what resampling is and why it is beneficial.
- Understand how to extend SMC to any diagram of probability kernels and weights.
- Know why sample impoverishment can occur in general models.
- Understand how MCMC moves work and why they increase diversity.

9. Variational Inference

- Understand the basic idea of VI and why it can be achieved by optimizing the ELBO.
- Know at least two common variational families.
- Understand the differences between MCMC and VI in (Num)Pyro (9.2).

Exercises

Exercise 1

- Know how to estimate expectations using Monte Carlo simulation, including (conditional) probabilities.
- Understand how this relates to rejection sampling.

Exercise 2

- Understand the workflow of building and implementing a graphical model.
- Know how to use the resulting program to estimate quantities of interest.

Exercise 3

- Understand the naive Bayes model, why it is attractive computationally, and its limitations.

Exercise 4

- Understand the forward algorithm for HMMs conceptually as composition.

Exercise 5

- Be able to translate a generative model into a `WeightedSampling.torch` model using `sample`, `deterministic`, and `observe`.
- Understand how these primitives relate to the weighted sampling procedure in the conditioning chapter.

Exercise 6

- Understand the interpretation of linear regression coefficients.
- Understand why logistic regression coefficients are not directly interpretable in the same way.
- Know how to nonetheless probe the implications of a fitted model (posterior predictions).

Exercise 7

- Know the basic workflow of MCMC in NumPyro. You do not need to know all code details, but you should be able to explain the basic steps you would take during coding.

Exercise 8

- Understand the multilevel regression model.

Exercise 9

- Understand how complete observations split models.
- Understand how to fit discrete binary kernels using a closed-form formula (Task 2).